Red Hat Consulting

Consulting Engagement Report

Department of Defence - OpenShift Container Platform install and knowledge transfer.

Version 2024-04-26-03-39-20

Table of Contents

Chapter 1. Preface

1.1. Confidentiality, Copyright, and Disclaimer

This is a customer-facing document between Red Hat, Inc. and Department of Defence ("Client").

This document is not a quote and does not include any binding commitments by Red Hat. If acceptable, a formal quote can be issued upon request, which will include the scope of work, cost, and any customer requirements as necessary.

This Consulting Engagement Report (CER) contains both Red Hat Confidential Information and Client Confidential Information subject to the confidentiality terms of the agreement governing the Services provided by Red Hat.

1.2. Audience

This document is intended for Client technical staff responsible for the environment.

1.3. Additional Background and Related Documents

This document does not contain step by step details of installation or other tasks, as they are covered in the relevant documentation on http://access.redhat.com/. Links to the appropriate documents will be made when required.

1.4. Scripts and playbooks

Any scripts provided are being provided as-is, without any form of support or warranty.
All provided scripts can be modified by the customer at will.

1.5. Customer satisfaction survey (CSat)

Red Hat is committed to a quality customer experience and continuous improvement in our services. Your feedback is the most important factor in determining if we've met your expectations and what we can do to improve. You will be receiving a customer satisfaction survey at the conclusion of our engagement. Please take just a few minutes to provide us with an honest snapshot of your experience. In the meantime, if there are any areas of concern, issues or problems on the project, please don't hesitate to reach out to Andrew Lloyd.

Chapter 2. Project Information

2.1. Originator

Red Hat Consulting

2.2. Owner

Red Hat Services

2.3. Document Conventions

2.3.1. Admonitions and Call-Outs

Additional detail about a topic will be called out with the "info" symbol.

Tips and advice will appear with a "lightbulb" admonition.

Warnings and strong recommendations will appear next to a caution exclamation.

Crucial advice and information will be appear next to a red exclamation.

2.3.2. Single-line Commands and Code

If instructed to type a single command, it will be displayed in a monospace font and highlighted like this example:

`ipa -vvv cert-find --all`

2.3.3. Multi-line Commands, Code / File Contents

If instructed to type multi-line commands, create or modify code blocks and or file contents the information will be monospaced and placed in a callout box like this example:

```
require 'sinatra'

get '/hi' do
  "Hello World!"
end
```

2.4. Additional Copies

To request additional copies of this Consulting Engagement Report, please contact the Red Hat Project Manager identified in Participants of the Engagement section or your current Red Hat account team.

2.5. Participants of the Engagement

2.5.1. Red Hat

Table 1. Red Hat Participants

Name	Role	E-mail address
Andrew Lloyd	Project Manager	alloyd@redhat.com
Craig Scott	Senior Consultant	craig.scott@redhat.com

2.5.2. Department of Defence

Table 2. AIR3503 Participants

Name	Role	E-mail address
Hashantha Mendis	Project manager	hashantha.mendis@defence.gov.au
Christian Townsend	Architect	christian.townsend@defence.gov.au
Leigh Grgurovic	IT Operations Manager	leigh.grgurovic@defence.gov.au
Joanne Makela	Procurement Officer	joanne.makela@defence.gov.au

Chapter 3. Executive Summary

Red Hat Consulting was engaged by the Defence AIR3503 project to build an OpenShift Container Platform environment to provide a platform for the test and development of containerised applications.

Chapter 4. Overview

4.1. About AIR3503

The Department of Defence is the Australian governments military. Defence's primary role is to defend Australia and its national interests, promote security and stability, and support the Australian community as directed by the Government.

4.2. Documentation Provided By AIR3503

Documentation Provided by Department of Defence

None provided.

4.3. Purpose And Engagement Approach

Red Hat Consulting was engaged by AIR3503 to assist with OpenShift Container Platform install and knowledge transfer. which seeks to:

- Review current design documentation.
- Install and configure Red Hat OpenShift over the VMware vSphere.
- Install and configure other software apps like Red Hat Ansible where required.
- Automate provisioning using Gitlab and other methods (as required).
- Provide SME knowledge to the broader build team on design improvements.
- Document design changes as required with respect to the OpenShift deployment.
- Prepare as-built drawings and documents based on the approved design.

4.4. Scope Summary

This section details Red Hat's strategy to assist AIR3503 with meeting their goals for this engagement. The following table outlines the focus of each phase.

Table 3. Engagement Scope Summary

Sprint	Week	Objectives
1	TODO	• TODO • TODO
	TODO	• TODO • TODO

Sprint	Week	Objectives
2	TODO	• TODO • TODO
	TODO	• TODO • TODO

1) Review current design documentation.

2) Install and configure Red Hat OpenShift over the VMware vSphere.

3) Install and configure other software apps like Red Hat Ansible where required.

4) Automate provisioning using Gitlab and other methods (as required).

5) Provide SME knowledge to the broader build team on design improvements.

6) Document design changes as required with respect to the OpenShift deployment.

7) Prepare as-built drawings and documents based on the approved design.

4.5. Engagement Requirement Criteria

Requirement	Details
Review current design documentation.	• Review current design documentation. • Provide recommendations on possible updates to design
Install and configure Red Hat OpenShift over the VMware vSphere.	• Mirror install software and images into environment • Build VMs • Build OpenShift Container Platform cluster
Install and configure other software apps like Red Hat Ansible where required.	• Install GitLab • Install Ansible Runner to GitLab
Automate provisioning using Gitlab and other methods (as required).	• Build OpenShift configuration repository • Build Ansible playbooks to apply configuration using GitLab Runner
Provide SME knowledge to the broader build team on design improvements.	• Ad-hoc knowledge transfer sessions

Requirement	Details
Document design changes as required with respect to the OpenShift deployment.	• Document design changes as required with respect to the OpenShift deployment.
Prepare as-built drawings and documents based on the approved design.	• Prepare as-built drawings and documents based on the approved design.

Chapter 5. Navigate

Navigate is a Red Hat® Consulting experience-driven framework that helps customers understand, plan, and move toward the successful adoption of Red Hat technologies.

Regardless of the current implementation stage of a modernization project, a Navigate engagement aligns a customer's existing and future digital transformation goals with Red Hat technologies.

Navigate begins with an assessment that establishes customer context in the areas of business strategy and technology landscape. Experienced Red Hat architects then run a series of workshops covering technical and business topics to help customers realize the best value from Red Hat products and technologies.

A Navigate engagement provides ways for customers to use Red Hat open source technology to achieve their defined business priorities and goals. Red Hat's architects and consultants help clients create a phased, sprint-based engagement to develop a foundational platform as a proving ground for IT capabilities and operations.

Read more about Red Hat Consulting offering: Navigate.

Chapter 6. Implementation Details

6.1. Journal

Week ending	Activities	Blockers
22/03/2024	• Onboarding	
29/03/2024	• Download OCP software and container images • Installed GitLab software • Enabled GitLab container registry	• Absence of a Linux host on the low side for mirroring OpenShift content
19/04/2024	• GitLab driven configuration of OpenShift with GitLab Runner • OpenShift installation complete but with VSphere storage issues, project will be purchasing a SAN so this may not be a issue • OpenShift configuration almost complete	• Absence of a Linux host on the low side for mirroring OpenShift content • Need CA to issue certificates • Need SAN (CSI compatible)
26/04/2024	• OpenShift installation complete without persistent storage • OpenShift configuration complete via GitLab • Hand-over documentation in progress	• Absence of a Linux host on the low side for mirroring OpenShift content • Need CA to issue certificates • Need SAN (CSI compatible)

6.2. Deployed Architecture

This content describes the architecture of the deployed and affected systems during this engagement.

6.2.1. Diagrams

Pictorial representations of the deployed systems.

TODO - insert logical, schematic, detail, and workflow diagrams representing the deployed and related/relevant systems

6.2.2. Processes

Information about the processes for using the deployed and related systems.

TODO

6.2.3. Environment Setup

Steps to prepare or create the environment for the deployed architecture.

TODO

6.2.4. Quality of Service Strategy

The following considerations were made to ensure enterprise ready uptime.

6.2.4.1. High Availability

TODO

6.2.4.2. Failover

TODO

6.2.4.3. Disaster Recovery

TODO

6.2.5. Service Levels

6.2.5.1. Service Level Indicators

TODO

6.2.5.2. Service Level Objectives

TODO

6.2.5.3. Service Level Agreements

TODO

6.2.6. TLS/SSL Certificates

The below table provides details on the TLS/SSL certificates that have been deployed as part of the engagement. To avoid an interruption in service ensure that the certificates are renewed before the expiration date.

TODO - Complete table with TLS/SSL certificate details

Table 4. TLS/SSL Certificates Information

Application	Certificate Location		Certificate Authority	Expiration Date
Example App	Server	*app.example.com*	*self-signed*	*12/31/2023*
	Certificate	*/etc/app/app.cert*		
	Key	*/etc/app/app.key*		

6.2.7. Observability

Information about how deployed systems can be observed.

6.2.8. Architectural Records

Information that affects how and why the deployed system was deployed the way it was.

6.2.8.1. Policies

Architectural policies are statements of *how things will be done*, typically legally binding and imposed from outside the organization. The policies documented here were provided throughout the course of the engagement from various teams and key stakeholders and affected the documented Architectural Decisions.

Table 5. TODO a title for this policy record

ID	POLICY-TODO unique number
Title	TODO a title for this policy record
Description	TODO

Date in Force	TODO date this policy is in force, ex, current, planned for X date
Source	TODO external groups, law, or internal team who says this policy must be enforced, might be multiple levels, ex Chief Security Officer due to Federal Memorandum XYZ
Related Policies	• TODO list links to policies that this record is related to in some way and the way in which, they are related. EX: other policy acts in cooperation with this policy.
Related Principles	• TODO list links to policies that this record is related to in some way and the way in which, they are related. EX: principle is a result of this policy.
Related Decisions	• TODO list links to policies that this record is related to in some way and the way in which, they are related. EX: decision made is due in part to this policy

6.2.8.2. Principles

Architectural principles are rules or guidelines at the organizational or team level used to make architectural decisions. The principles documented here were provided throughout the course of the engagement from various teams and key stakeholders and affected the documented Architectural Decisions.

Table 6. TODO a title for this principle record

ID	PRINCIPLE-TODO unique number
Title	TODO a title for this principle record
Description	TODO long form of what the principle is
Motivation	TODO why was this principle put in place, ex no deployments on fridays, because they had a history of bad deployments causing people to work all weekend
Source	TODO team name or key stake holder name who provided the applicable principle
Related Policies	• TODO list links to policies that this record is related to in some way and the way in which, they are related. EX: policy is reason for this principle
Related Principles	• TODO list links to policies that this record is related to in some way and the way in which, they are related. EX: other principle works in cooperation with this principle
Related Decisions	• TODO list links to policies that this record is related to in some way and the way in which, they are related. EX: decision made is due in part to this principle

6.2.8.3. Decisions

Architectural decisions made throughout the course of the engagement with information on the investigated alternatives and justification for the decisions made.

Table 7. TODO Title

<table>
<tr><td>ID</td><td>ADR-TODO unique number</td></tr>
<tr><td>Title</td><td>TODO a title for this decision record</td></tr>
<tr><td>Status</td><td>TODO Set or update the decision’s status</td></tr>
<tr><td>Subject Area(s)</td><td>TODO subject areas, can think of this like hashtags</td></tr>
<tr><td>Question</td><td>TODO State the to-be decision as a question</td></tr>
<tr><td>Issue or Problem</td><td>TODO Context for why the architectural question is being asked.</td></tr>
<tr><td>Assumptions</td><td>TODO What is believed to be true about the context of the problem, constraints on the solution, and so on.</td></tr>
<tr><td>Alternatives</td><td>TODO An enumerated list of alternatives and explanations.</td></tr>
<tr><td>Decision</td><td>TODO The decision taken. This should be one of the named Alternatives; for example: “Alternative 2 is chosen.”</td></tr>
<tr><td>Justification</td><td>TODO Why the decision was made</td></tr>
<tr><td>Agreeing Parties</td><td><table>
<tr><th>Person</th><th>Representing</th></tr>
<tr><td>TODO Agreeing person name</td><td>TODO Team or group that person is representing and agreeing on behalf of, ex security, operations</td></tr>
<tr><td>TODO Agreeing person name</td><td>TODO Team or group that person is representing and agreeing on behalf of, ex security, operations</td></tr>
</table></td></tr>
<tr><td>Related Policies</td><td>• TODO list links to policies that this record is related to in some way and the way in which, they are related. EX: policy affected decision in some way.</td></tr>
<tr><td>Related Principles</td><td>• TODO list links to policies that this record is related to in some way and the way in which, they are related. EX: principle affected decision in some way.</td></tr>
<tr><td>Related Decisions</td><td>• TODO list links to policies that this record is related to in some way and the way in which, they are related. EX: other decision limited options for this decision</td></tr>
</table>

6.2.9. Red Hat OpenShift Container Platform 4 - VMware vSphere Architecture

All OpenShift environments for on prem environments will utilize hardware to achieve some level of redundancy. Each master and infrastructure component (EFK, metrics, router, and registry) should reside on different physical hardware. The image below illustrates a simple cluster example.

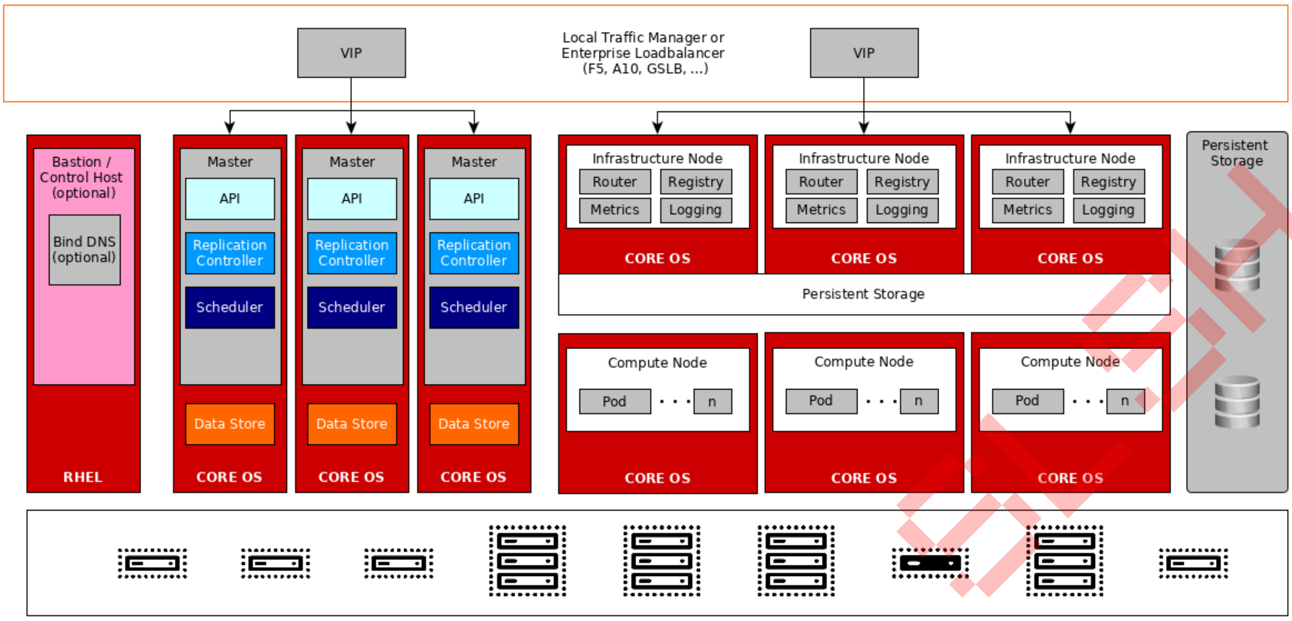

Figure 1. Cluster Diagram

6.3. Technical Implementation

TODO

6.3.1. Red Hat OpenShift Container Platform 4 - VMware vSphere

6.3.1.1. TODO_ENVIRONMENT_NAME

TODO Environment Description

6.3.1.1.1. Domain Information

Domain: `n/a`
Cluster Name: `n/a`

6.3.1.1.2. VMware Information

The following VMware data has been configured in the OpenShift cluster:

Table 8. VMware Data

VMware Property	Value
VCENTER_SERVER	vcsa.rhbr-labs.com
VCENTER_SERVER	vcsa.rhbr-labs.com
VCENTER_USER	administrator@vsphere.local
VCENTER_PASSWORD	<hidden>
VCENTER_DC	rhbr-labs-dc
VCENTER_DS	rhbr-labs-ds

6.3.1.1.3. Network Information

Table 9. Network Data

Name	CIDR Block or Value	Comments
VNet		
Pod Network	/16	IP Addresses
Service Network	/16	IP Addresses

Name	CIDR Block or Value	Comments
Host Subnet Length	9	This tells OpenShift to allocate a /23 (32-9) of [Pod Network] to every node. NOTE: The default installation setting is a subnet of /9. This value must be reduced for smaller CIDR ranges.

6.3.1.1.4. Node Information

Table 10. Nodes Data

Server FQDN	IP	Role	Subs Used	Labels
		Master	N/A	TODO
		Master	N/A	TODO
		Master	N/A	TODO
		Infra	N/A	TODO
		Infra	N/A	TODO
		Infra	N/A	TODO
		Application	1	TODO
		Application	1	TODO
		Application	1	TODO
		Bastion	N/A	N/A

6.3.1.1.5. Storage Information

Table 11. Storage Data

Application	Storage Type	Backing Storage	Size
Metrics	Block		
Logging	Block		
Registry	File		
Application	File		

6.4. Validation Procedure

TODO

6.4.1. Red Hat OpenShift Container Platform 4 - VMware vSphere

6.4.1.1. Pre Deployment Environment Verification Checklist

Prior to or as part of the engagement the "Openshift Container Platform (OCP) 4 Prerequisite Checklist" document was delivered.

As part of the engagement that checklist was filled out per deployed environment. Those external documents can be referenced here: * TODO_environment_1_name - TODO_link_or_describe_location_of_completed_checklist_for_environment_1 * TODO_environment_2_name - TODO_link_or_describe_location_of_completed_checklist_for_environment_2

It is recommended for any future OpenShift 4.x clusters that are created that the same check list be followed.

6.4.1.2. Post Deployment Environment Verification Checklist

6.4.1.2.1. Environment TODO_ENV_NAME verification (TODO: copy this per cluster)

Table 12. Cluster Durability and Recovery

Requirement	Result
Plan for Node, Rack, or AZ Failure	TODO
Plan for Data Center/Region Failure	TODO
Master Anti-Affinity/Failure Domains	TODO
Infrastructure Anti-Affinity/Failure Domains	TODO
Cluster Capacity Planning Completed	TODO
Cluster Capacity for Failing Nodes/Racks	TODO
Defined SLA for OCP Cluster: Does deployed architecture support planned SLA?	TODO

Requirement	Result
Documented SLAs for Dependencies. Should cover: • Underlying Compute, Network, Storage • External Load Balancer • External Registry • Artifact Repositories • CICD (ie Jenkins/Tower/etc)	TODO
If MultiCluster Failover: • Failover Process Documented and Tested • Recovery Process Documented and Tested	TODO
Etcd Backup Configured	TODO
Etcd Restore Documented and Tested	TODO
If Required, Infra Backup Configured and Tested. Should cover: • Logging ElasticSearch • Prometheus Metrics	TODO
Disaster Recovery Procedure Documented and Tested	TODO

Table 13. Configuration and Durability

Requirement	Result
Requests and Limits Defined According to Desired QoS	TODO
Liveness and Readiness Probes Defined and Correct: • Liveness probe does not test for downstream service health	TODO
Connections and requests to downstream services are retried on failure or gracefully handled	TODO
Repeated failures by downstream services causes future requests to be short circuited via a circuit breaker	TODO
Application gracefully shuts down on SIGTERM • Incoming requests are handled during grace period • Idle keep-alive sockets are closed	TODO
Load Test Completed: * What is capacity at maximum allowed latency in a single cluster?	TODO

Requirement	Result
Functional Testing Completed	TODO
If Required by SLA, Deployments Scaled to Multiple Replicas	TODO
Pod Disruption Budget Defined	TODO

Table 14. Security

Requirement	Result
ResourceQuotas are Defined for All Application Projects	TODO
Pods Configured with Most Restrictive Possible SCC	TODO
Narrowly Defined NetworkPolicy Exists for all Pods	TODO
User Access to Namespaces Audited	TODO
`ovs-networkpolicy` minimumally configured for project isolation. See the following for suggested implementation details: • redhat-cop/openshift-toolkit/networkpolicy • Blog - NetworkPolicies and Microsegmentation	TODO

Table 15. Observability

Requirement	Result
Application Logs to stdout, Logs Aggregated	TODO
If Using Prometheus, ServiceMonitors and Rules Defined	TODO

6.5. Knowledge Transfer

TODO

Table 16. Working Sessions

Session	Date	Purpose / Attendees	Outputs
TODO - session title	TODO - date	• Purpose ◦ TODO ◦ TODO • Attendees ◦ TODO ◦ TODO	• TODO - outputs • TODO - outputs

6.6. Challenges, Resolutions, and Recommendations

6.6.1. TODO_sample_challenge_name_1

6.6.1.1. Challenge

TODO

6.6.1.2. Immediate Resolution

TODO

6.6.1.3. Future Recommendation

TODO

Chapter 7. Recommendations

7.1. Technical Next Steps

TODO

7.2. Relevant Training

7.2.1. Red Hat Training

Specific course descriptions and availability can be found on Red Hat's website under Training and Certification. The following courses are recommended:

7.2.1.1. Red Hat Learning Subscription

Fill skills gaps and address business challenges by taking advantage of unlimited access to our comprehensive curriculum.

For full details, or to enroll:
https://www.redhat.com/en/services/training/learning-subscription#tab.rhls.1

7.2.1.2. Red Hat Skills Assessment

Use the skills assessment tool to discover what training opportunities you may benefit from.

For full details, or to enroll:
https://www.redhat.com/rhtapps/assessment

7.2.1.3. Platform Courses

7.2.1.3.1. Red Hat System Administration I (RH124)

Course description

The first of two courses covering the core system administration tasks needed to manage Red Hat Enterprise Linux servers

Red Hat System Administration I (RH124) is designed for IT professionals without previous Linux system administration experience. The course provides students with Linux administration competence by focusing on core administration tasks. This course also provides a foundation for students who plan to become full-time Linux system administrators by introducing key command-line concepts and enterprise-level tools.

This course is the first of a two-course series that takes a computer professional without Linux system administration knowledge to become a fully capable Linux administrator. These concepts are further developed in the follow-on course, Red Hat System Administration II (RH134).

This course is based on Red Hat® Enterprise Linux 9.0.

Following course completion, you will receive a 45-day extended access to hands-on labs for any course that includes a virtual environment.

Note: This course is offered as a five day in person class, a five day virtual class or is self-paced. Durations may vary based on the delivery. For full course details, scheduling, and pricing, select your location then "get started" on the right hand menu.

Course content summary

- Introduce Linux and the Red Hat Enterprise Linux ecosystem.
- Run commands and view shell environments.
- Manage, organize, and secure files.
- Manage users, groups and user security policies.
- Control and monitor systemd services.
- Configure remote access using the web console and SSH.
- Configure network interfaces and settings.
- Manage software using DNF

Audience for this course

- The primary persona is a technical professional with current or pending responsibilities as a Linux enterprise or cloud system administrator.
- This course is popular in Red Hat Academy, and is targeted at the student who is planning to become a technical professional using Linux.

Prerequisites for this course

- Basic technical user skills with computer applications on some operating systems are expected.
- Take our free assessment to gauge whether this offering is the best fit for your skills.

Technology considerations

- There are no special technical requirements for this course.
- This course is not designed for bring your own device (BYOD).
- Internet access is not required but is recommended so that students can research and browse Red Hat online resources.

For full details, or to enroll: https://www.redhat.com/en/services/training/rh124-red-hat-system-administration-i

7.2.1.3.2. Red Hat System Administration II (RH134)

Course Description

Build the skills to perform the key tasks needed to become a full-time Linux administrator

Red Hat System Administration II (RH134) is the second part of the RHCSA training track for IT professionals who have already attended Red Hat System Administration I. The course goes deeper into core Linux system administration skills in storage configuration and management, installation and deployment of Red Hat Enterprise Linux, management of security features such as SELinux, control of recurring system tasks, management of the boot process and troubleshooting, basic system tuning, and command-line automation and productivity. This course assumes that students have attended Red Hat System Administration I (RH124).

Experienced Linux administrators who seek rapid preparation for the RHCSA certification should instead start with RHCSA Rapid Track (RH199).

This course is based on Red Hat Enterprise Linux 9.0.

Following course completion, you will receive a 45-day extended access to hands-on labs for any course that includes a virtual environment.

Note: This course is offered as a five day in person class, a five day virtual class or is self-paced. Durations may vary based on the delivery. For full course details, scheduling, and pricing, select your location then "get started" on the right hand menu.

Course summary

- Install Red Hat Enterprise Linux using scalable methods
- Access security files, file systems, and networks
- Execute shell scripting and automation techniques
- Manage storage devices, logical volumes, and file systems
- Manage security and system access
- Control the boot process and system services
- Run containers

Audience for this course

This course is geared toward Windows system administrators, network administrators, and other system administrators who are interested in supplementing current skills or backstopping other team members, in addition to Linux system administrators who are responsible for these tasks:

- Configuring, installing, upgrading, and maintaining Linux systems using established standards and procedures
- Providing operational support
- Managing systems for monitoring system performance and availability
- Writing and deploying scripts for task automation and system administration

Recommended training

- Successful completion of Red Hat System Administration I (RH124) is recommended.
- Experienced Linux administrators seeking to accelerate their path toward becoming a Red Hat Certified System Administrator should start with the RHCSA Rapid Track course (RH199).
- Take our free assessment to gauge whether this offering is the best fit for your skills.

Technology considerations

- There are no special technology requirements.
- This course is not designed for bring your own device (BYOD).
- Internet access is not required, but is recommended to allow students to research and access Red Hat online resources.

For full details, or to enroll: https://www.redhat.com/en/services/training/rh134-red-hat-system-administration-ii

7.2.1.4. Ansible Courses

7.2.1.4.1. Red Hat Enterprise Linux Automation with Ansible (RH294)

Course description

Learn how to automate Linux system administration tasks with Red Hat Ansible Automation Platform

Red Hat Enterprise Linux Automation with Ansible (RH294) is designed for Linux administrators and developers who need to automate repeatable and error-prone steps for system provisioning, configuration, application deployment, and orchestration.

This course is based on Red Hat® Enterprise Linux® 9 and Red Hat Ansible Automation Platform 2.2.

Following course completion, you will receive a 45-day extended access to hands-on labs for any course that includes a virtual environment.

Note: This course is offered as a four day in person class, a five day virtual class or is self-paced. Durations may vary based on the delivery. For full course details, scheduling, and pricing, select your location then “get started” on the right hand menu.

Course content summary

- Install Red Hat Ansible Automation Platform on control nodes.
- Create and update inventories of managed hosts and manage connections to them.
- Automate administration tasks with Ansible Playbooks and ad hoc commands.
- Write effective playbooks at scale.
- Protect sensitive data used by Ansible Automation Platform with Ansible Vault.
- Reuse code and simplify playbook development with Ansible Roles and Ansible Content Collections.

Audience for this course

This course is geared toward Linux system administrators, DevOps engineers, infrastructure automation engineers, and systems design engineers who are responsible for these tasks:

- Automating configuration management
- Ensuring consistent and repeatable application deployment
- Provisioning and deployment of development, testing, and production servers
- Integrating with DevOps continuous integration/continuous delivery workflows

Prerequisites for this course

- Pass the Red Hat Certified System Administrator (RHCSA) exam (EX200), or demonstrate equivalent Red Hat Enterprise Linux knowledge and experience.
- Take our free assessment to gauge whether this offering is the best fit for your skills.

For full details, or to enroll: https://www.redhat.com/en/services/training/rh294-red-hat-linux-automation-with-ansible

7.3. Other

TODO

Chapter 8. Post-Engagement Risks

8.1. Scalability and Supportability Risks

TODO

Appendix A: Glossary

Table 17. General Terminology

Term	Definition
BZ	Bugzilla. This context refers to the specific instance of Bugzilla operated by Red Hat to track bugs, RFEs, and release information. Also, it may refer to a specific numbered Bugzilla bug when given with a definite article or number.
Case	Support Case, RH Case,A Red Hat support case, requested and prioritized by the client, then responded to directly by GSS staff. This case number is distinct from a BZ bug number (most RH support cases do not have a corresponding BZ and vice-versa).
DNS	Domain Name Service
Downstream	Refers toward “productized” deployable, supported artifact (signed binary) end of open source software development process and related contracted errata, event-based support artifacts, product engineering, and product-specific documentation.
FQDN	Fully Qualified Domain Name
GSS (a.k.a. CEE)	Red Hat Global Support Services (“support”) aka “Customer Experience and Engagement”. Able to resolve bugs, configuration problems, and shepherd RFEs.
KB	Knowledge Base article, a short and specific how-to guide available in the customer portal.
RFE	Request for Enhancement. These are requests from clients or RH engineering for new product functionality or behavior. Generally tracked internally as a BZ.
RHC	Red Hat Consulting. Able to provide on-site and remote engineering and analysis for solution creation.
RHEL	Red Hat Enterprise Linux
RPM	RPM Package Manager (a recursive acronym)
TAM	Technical Account Manager. A site-designated support resource manager from GSS that maintains site-specific knowledge, awareness, and drives resolution.
Upstream	Refers toward source-code origin end of open source software development process and related open source community collaboration, documentation, and testing.

Appendix B: Metrics

TODO

B.1. Perception and Proficiency

This section discusses how to capture the perception of OpenShift at the AIR3503 and the Department of Defence's proficiency at using and operating OpenShift, and the value of these key areas.

B.1.1. Perception

The most subjective and people-driven measurements and practices measure how the business views and promotes the platform according to individual contributors; we capture all of these values through surveys.

- **Perception: Value** - The measure of the technology's contribution to the Department of Defence business value.
- **Perception: Simplicity** - Measures the perceived usability, how easy it is to work with the technologies, and the perceived learning curve.
- **Perception: Trialability** - Measures the degree to which the technology may be experimented with on a limited basis (e.g., in a non-prod environment).
- **Perception: Compatibility** - Measures the perceived degree of consistency of the technology with the existing culture and the past experiences, and the established business objectives; this also includes reusability of platform compoments and processes.
- **Perception: Observability** - It measures how easy it is to showcase and promote platform usage/use-cases (results demonstrability).

B.1.2. Proficiency

Organization-driven measurements and practices that create business value through organizational knowledge and process governance.

- **# surveys completed** - Number of surveys completed
- **# teams onboarded** - Number of teams onboarded to OpenShift
- **# users onboarded** - Number of developers onboarded to OpenShift

B.2. Performance and Utilization

B.2.1. Performance

Platform-driven measurements and practices that align application, developer, and operational performance to customer business value and KPIs.

B.2.1.1. Common Performance Metrics

- **MTTR** - How long it takes to restore service when a service incident occurs.
- **Deployment Error Rate** - An essential quality metric that measures what percentage of changes to production fail. It is crucial to have alignment on what constitutes a failure. The recommended definition is a change that either results in degraded service or require remediation.
- **CPU Utilization** - The amount of processing power that a specific application is using on a computer's central processing unit (CPU). It is expressed as a percentage of the total available CPU resources and represents the amount of time that the CPU is actively working on tasks related to that specific application.
- **Memory utilization** - The amount of physical memory (RAM) that is being used by a computer system at a given point in time. It is expressed as a percentage of the total available memory, indicating how much of the memory is being used and how much is still available.

B.2.1.2. OpenShift Performance Metrics

- **lead-time to scale** - The time it takes to scale an application running in production.
- **lead-time to onboard users** - The time it takes to process and integrate new users onto the platform.
- **lead-time to onboard apps** - The time it takes to onboard the application to the platform.
- **lead-time for change** - The time it takes to go from code committed to code successfully running in production. Shorter lead times enable faster feedback on what we are building and encourage smaller batch sizes to increase flow.

B.2.2. Utilization

Measures how the business uses the solution in production.

B.2.2.1. OpenShift Utilization Metrics

- **workloads onboarded (prod)** - Number of applications running in OpenShift production.
- **workloads onboarded (LLE)** - Number of applications running in OpenShift lower level environment, e.g. Dev or Non-Prod.

B.3. Results

TODO

B.3.1. Topic: Perception

TODO

Metric	Baseline (during Navigate)	Final (engagement closeout)
Value	TODO	TODO
Simplicity	TODO	TODO
Trialability	TODO	TODO
Compatibility	TODO	TODO
Observability	TODO	TODO

B.3.2. Topic: Proficiency

TODO

Metric	Baseline (during Navigate)	Final (engagement closeout)
Score (from survey)	TODO	TODO
# surveys completed#	TODO	TODO
# teams onboarded	TODO	TODO
# users onboarded	TODO	TODO

B.3.3. Topic: Performance

TODO

Metric	Baseline (during Navigate)	Final (engagement closeout)
MTTR (minutes)	TODO	TODO
Deployment error rate	TODO	TODO
CPU utilization	TODO	TODO
Memory utilization	TODO	TODO
lead-time to scale (days)	TODO	TODO
lead-time to onboard users (days)	TODO	TODO
lead-time to onboard apps (days)	TODO	TODO
lead-time for change (promotion)	TODO	TODO

B.3.4. Topic: Utilization

TODO

Metric	Baseline (during Navigate)	Final (engagement closeout)
# workloads onboarded (prod)	TODO	TODO
# workloads onboarded (LLE)	TODO	TODO

Appendix C: Red Hat OpenShift Container Platform 4 - VMware vSphere

C.1. Deployment Procedure on VMware vSphere Documentation

This section contains the steps that were performed to deploy and configure the OpenShift cluster on VMware for the customer Department of Defence. The steps mentioned in this section are based on the following procedures from the official documentation:

- Installing on vSphere
- Configuring user authentication and access controls for users and services
- Ingress operator in OpenShift Container Platform
- Configuring registries for OpenShift Container Platform
- Configuring cluster logging in OpenShift Container Platform
- Configuring and using the monitoring stack in OpenShift Container Platform

C.1.1. Prerequisites Preparation

The following services are pre-requisites for OpenShift 4.14 installation on vSphere (see more on the official docs for OpenShift) :

- DHCP (If 4.14 >= 4.6 and you intend to provision nodes with UPI using static IPs DHCP isn't required)
- DNS
- Load Balancer
- Webserver (to host bootstrap ignition file)
- vSphere 6.7U3

C.1.2. Deploying OpenShift on VMware using User Provisioned Infrastructure (UPI)

The procedure used to deploy the OpenShift cluster on vSphere is a subset of the documentation found online at:

https://access.redhat.com/documentation/en-us/openshift_container_platform/4.14/html/installing/installing-on-vsphere
And
https://console.redhat.com/openshift/install/vsphere/user-provisioned

In order to deploy OpenShift on vSphere, the following steps needs to be performed.

C.1.2.1. Generate SSH key

See: Generating a key pair for cluster node SSH access

The specific command used for this engagement:

```
ssh-keygen -t rsa -b 2048 -f ~/.ssh/id_rsa
```

C.1.2.2. Download Binaries

See: Obtaining the installation program
And
See: Installing the OpenShift CLI by downloading the binary

The specific commands used for this engagement:

```
mkdir ~/bin

OCP4_BASEURL=https://mirror.openshift.com/pub/openshift-v4/clients/ocp/latest
LATEST_VERSION=$(curl -s ${OCP4_BASEURL}/release.txt | grep 'Version: ' | awk '{print $2}')
curl -s ${OCP4_BASEURL}/openshift-client-linux-$LATEST_VERSION.tar.gz | tar -xzf - -C ~/bin oc kubectl
curl -s ${OCP4_BASEURL}/openshift-install-linux-$LATEST_VERSION.tar.gz | tar -xzf - -C ~/bin/ openshift-install
```

C.1.2.3. Pull Secret

Pull secrets are specific each Red Hat account, so the customer was directed to log in to console.redhat.com and obtain a pull secret from the appropriate link at: https://console.redhat.com/openshift/install/vsphere/user-provisioned.

For this engagement the pull secret was stored in the file `~/ocp4_pull_secret`

C.1.2.4. Creating the Install Configuration File

See: Manually creating the installation configuration file

The installation configuration file was created using the following contents:

```
<replace values shown in this example with the actual values used>

mkdir ~/ocp4
cd ~/ocp4
```

```
export DOMAIN=rhbr-labs.com ①
export CLUSTERID=ocp ②
export VCENTER_SERVER=vcsa.rhbr-labs.com ③
export VCENTER_USER="administrator@vsphere.local" ④
export VCENTER_PASS='<admin_password>' ④
export VCENTER_DC='rhbr-labs-dc' ⑤
export VCENTER_DS='<datastore-id>' ⑥
export PULL_SECRET=$(cat ~/ocp4_pull_secret) ⑦
export OCP_SSH_KEY=$(cat ~/.ssh/id_rsa.pub) ⑧

cat <<EOF > install-config.yaml
apiVersion: v1
baseDomain: ${DOMAIN}
compute:
- hyperthreading: Enabled
  name: worker
  replicas: 1
controlPlane:
  hyperthreading: Enabled
  name: master
  replicas: 1
metadata:
  name: ${CLUSTERID}
networking:
  clusterNetworks:
  - cidr: 10.254.0.0/16 ⑨
    hostPrefix: 24
  networkType: OpenShiftSDN
  serviceNetwork:
  - 172.30.0.0/16 ⑩
platform:
  vsphere:
    vcenter: ${VCENTER_SERVER}
    username: ${VCENTER_USER}
    password: ${VCENTER_PASS}
    datacenter: ${VCENTER_DC}
    defaultDatastore: ${VCENTER_DS}
pullSecret: '${PULL_SECRET}'
sshKey: '${OCP_SSH_KEY}'
EOF
```

① The base domain of the cluster. All DNS records must be sub-domains of this base and include the cluster name.

② The cluster name that you specified in your DNS records.

③ The fully-qualified host name or IP address of the vCenter server.

④ vCenter credentials. Required privileges are detailed here: https://github.com/openshift/installer/blob/master/docs/user/vsphere/privileges.md.

⑤ The vSphere Datacenter.

⑥ Default Datastore to use.

⑦ Pull secret obtained in console.redhat.com.

⑧ The public portion of the default SSH key for the core user in Red Hat Enterprise Linux CoreOS (RHCOS).

⑨ Change this field if it is required to use a different cluster network range than the default one.

⑩ Change this field if it is required to use a different service network range than the default one.

> The installer will automatically delete the `install-config.yml` file when it is executed. It is strongly recommended that a backup be created prior to moving to the next step to avoid the need to re-create the file in the event it needs to be used again.

A backup of the installation file was created with the following command

```
cp install-config.yaml ../install-config.yaml.bkp
```

C.1.2.5. Create Manifests and Ignition Configuration

See: Creating the Kubernetes manifest and Ignition config files

The specific command used for this engagement (manifests are created in the current directory):

```
openshift-install create manifests
```

To ensure that masters are not schedulable the following command was used to modify the generated manifest files:

```
sed -i 's/mastersSchedulable: true/mastersSchedulable: false/g' manifests/cluster-scheduler-02-config.yml
```

The Ignition configuration was created using the following commands:

```
openshift-install create ignition-configs

cat <<EOF > append-bootstrap.ign
{
  "ignition": {
    "config": {
```

```
        "append": [
          {
            "source": "http://WEBSERVERIP:8080/ocp/ignition/bootstrap.ign",
            "verification": {}
          }
        ]
      },
      "timeouts": {},
      "version": "2.1.0"
    },
    "networkd": {},
    "passwd": {},
    "storage": {},
    "systemd": {}
  }
EOF
```


C.1.2.6. Creating the Virtual Machines on vSphere

See: Creating Red Hat Enterprise Linux CoreOS (RHCOS) Machines in vSphere

The specific commands used for this engagement to copy `bootstrap.ign` to the webserver:

```
sudo mkdir -p /var/www/html/ocp/ignition/
sudo cp bootstrap.ign /var/www/html/ocp/ignition/
```

This command confirms that webserver is hosting the bootstrap ignition file and that it is accessible:

```
curl http://WEBSERVERIP:8080/ocp/ignition/bootstrap.ign
```

This command was used to generate files in base64:

```
for i in append-bootstrap master worker
do
base64 -w0 < $i.ign > $i.64
done
```

C.1.2.6.1. Import OVA with vSphere

Access the vCenter web UI:
<replace this with the actual vCenter web URL> https://customer.vcenter.url.com

> You will need to connect to vSphere using credentials with privileges to create/upload templates to the target datacenter.

Click the icon resembling a stack of paper to navigate to “VMs and Templates”.

From there right click on your datacenter and select "**New Folder → New VM and Template Folder**".

Name this new folder the name of your cluster id: n/a

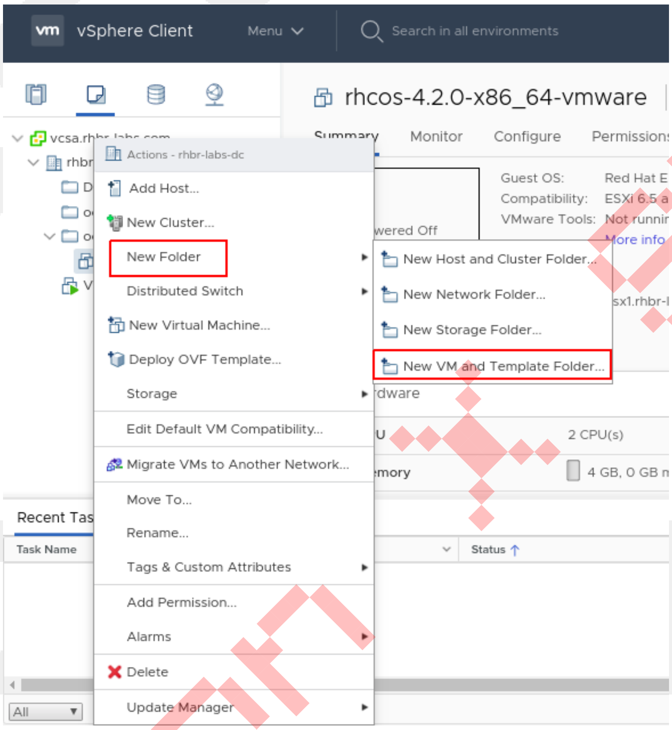

Figure 2. Creating Folder

Import the OVA by right clicking the folder and selecting “**Deploy OVF Template**”.

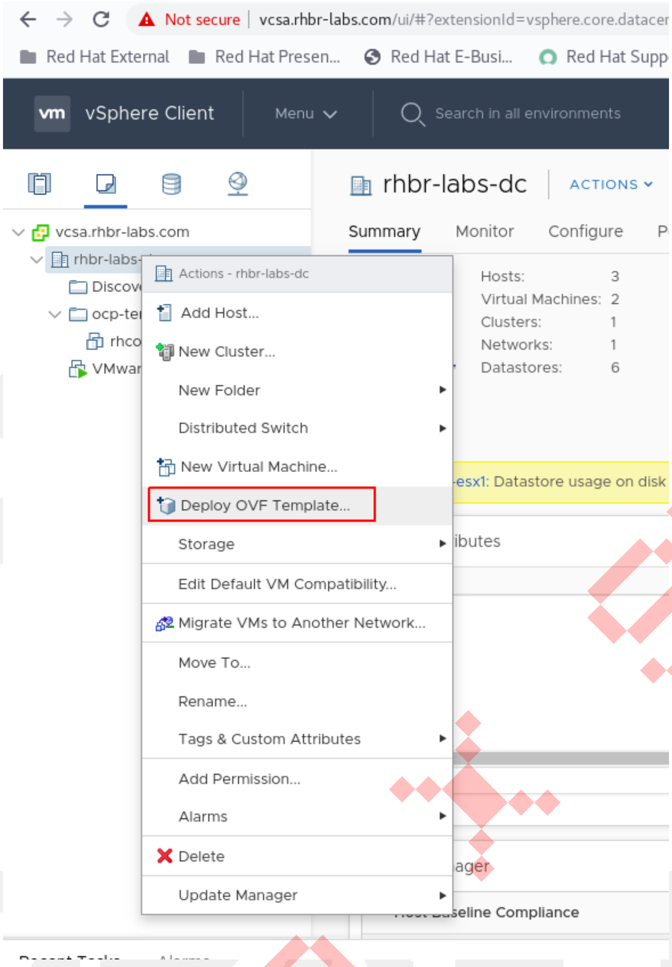

Figure 3. Deploying OVA

Add the url to RHCOS OVA (see here) and click "**NEXT**":

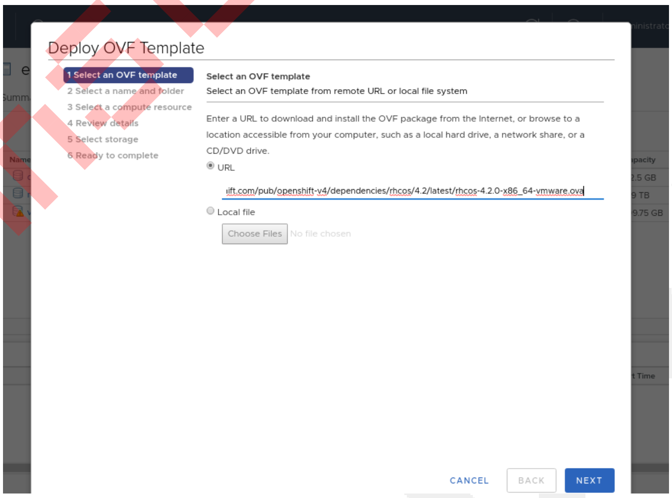

Figure 4. Deploying OVA

Select the folder you created in the previous step and click "**NEXT**":

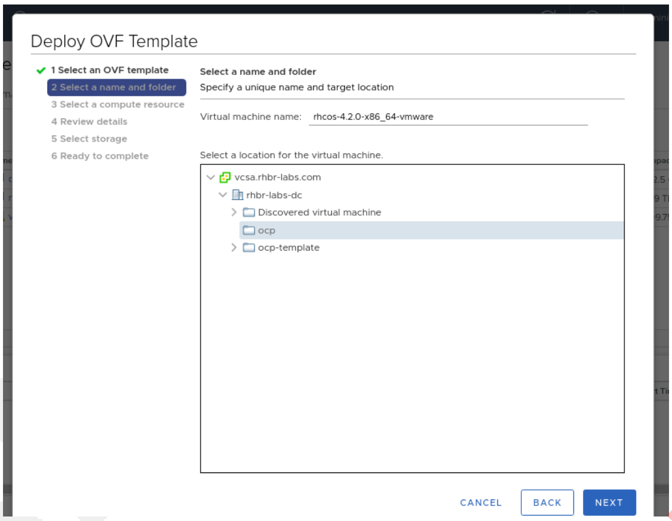

Figure 5. Deploying OVA

Select the compute resource and click "**NEXT**":

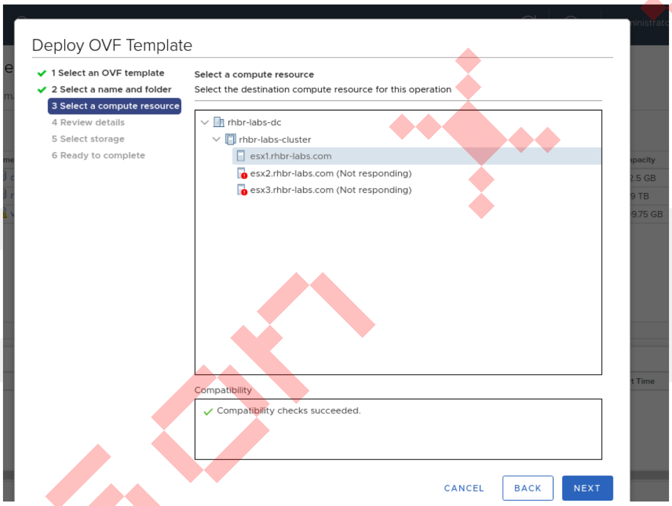

Figure 6. Deploying OVA

Select the datastore specified in the installation config file earlier:

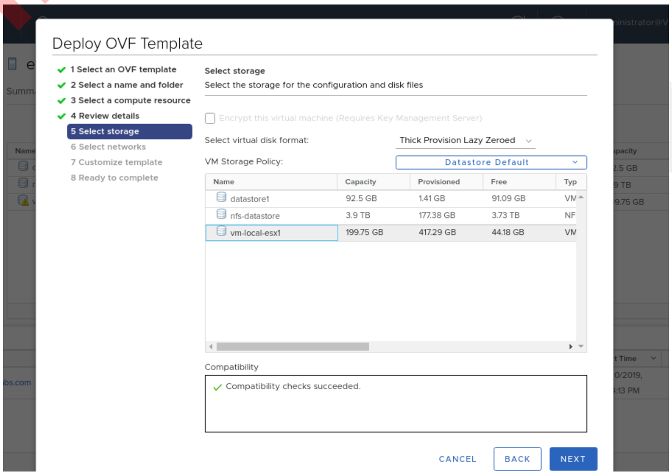

Figure 7. Deploying OVA

Select the network and click "**NEXT**":

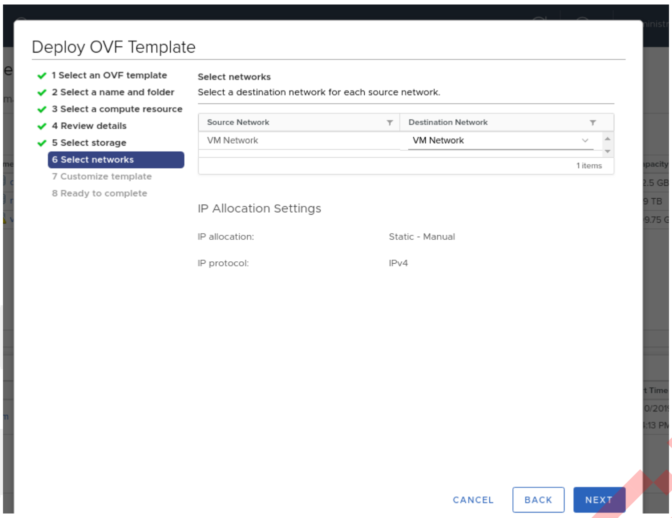

Figure 8. Deploying OVA

Don’t fill anything yet (these parameters will be filled further). Click "**NEXT**".

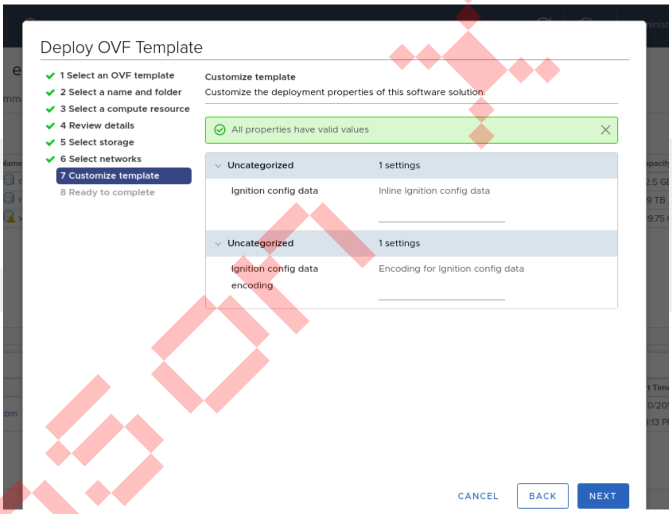

Figure 9. Deploying OVA

Click "**FINISH**" in the next screen

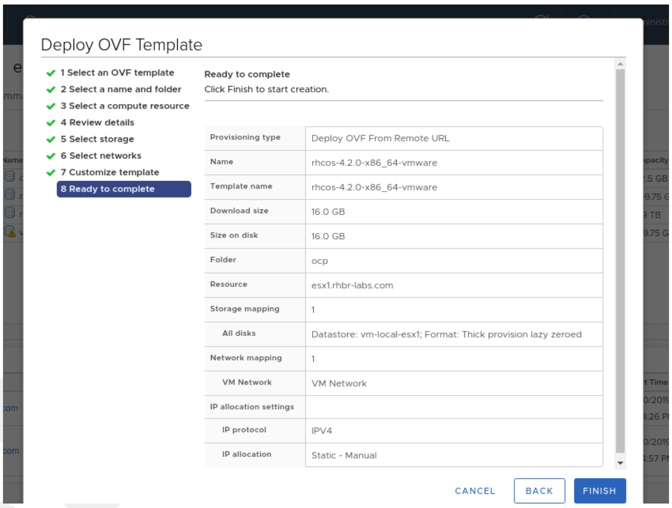

Figure 10. Deploying OVA

NEVER start up the template itself. The ignition process only runs on first boot, so booting the template would cause ignition files provided afterward to be ignored.

C.1.2.7. Provision OpenShift Servers

Right click on the OVA and select **Clone** → **Clone to Virtual Machine**

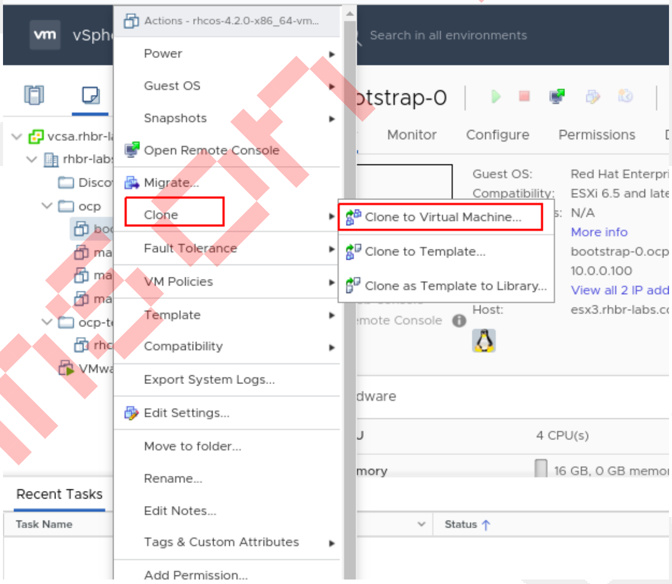

Figure 11. Clone to VM

Select the folder you created before, input the VM name and click "**NEXT**".

The VM name must match the name configured in DHCP and DNS.

```
Folder: n/a
VM Name: bootstrap-0
```

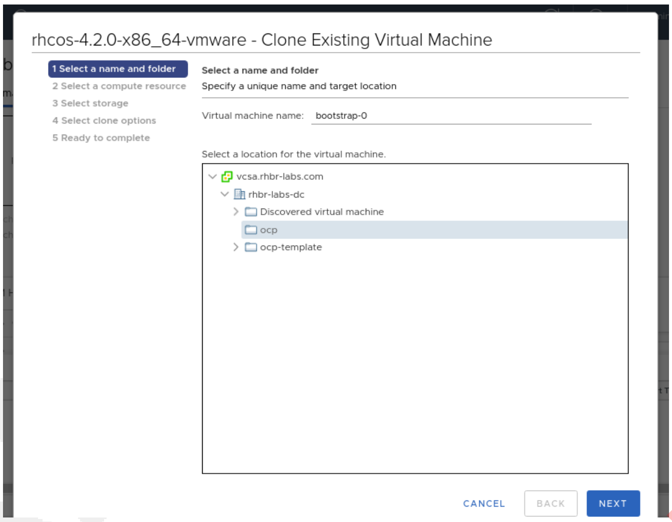

Figure 12. Clone to VM

Select the compute resource and click "**NEXT**":

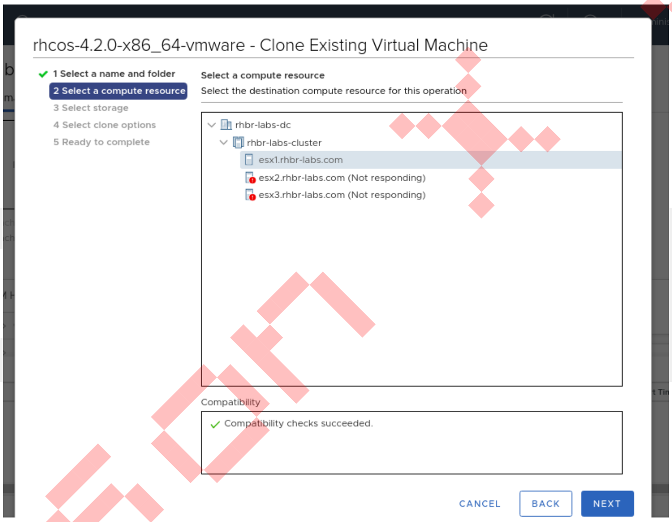

Figure 13. Clone to VM

Select the datastore and select disk format as "**Thin Provision**":

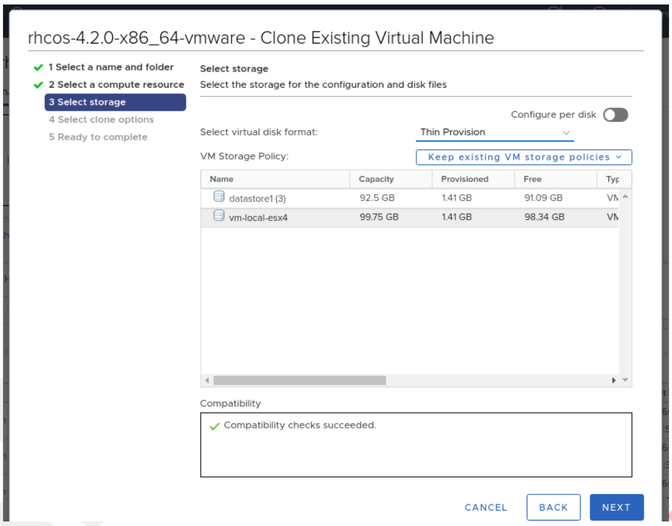

Figure 14. Clone to VM

Enable the option "**Customize this virtual machine's hardware**"

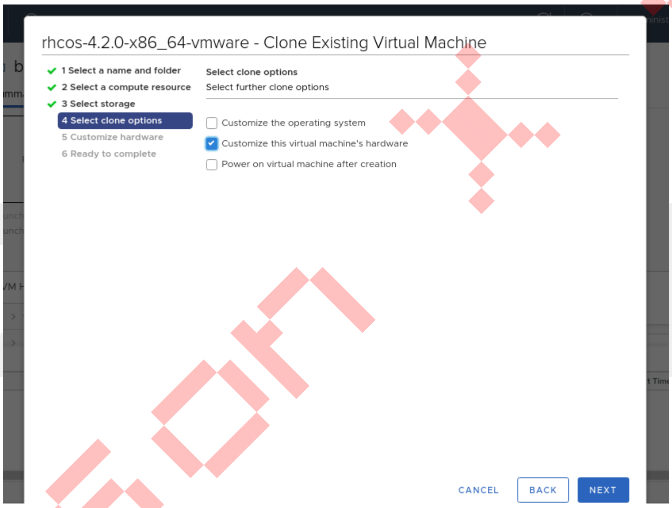

Figure 15. Clone to VM

In the next screen input the following parameters:

```
CPU: n/a
Memory: n/a
- Enable "Reserve all guest memory" option
Hard Disk: n/a
Network Adapter 1:
- MAC Address: Manual - <MAC ADDRESS RESERVED IN DHCP>
```

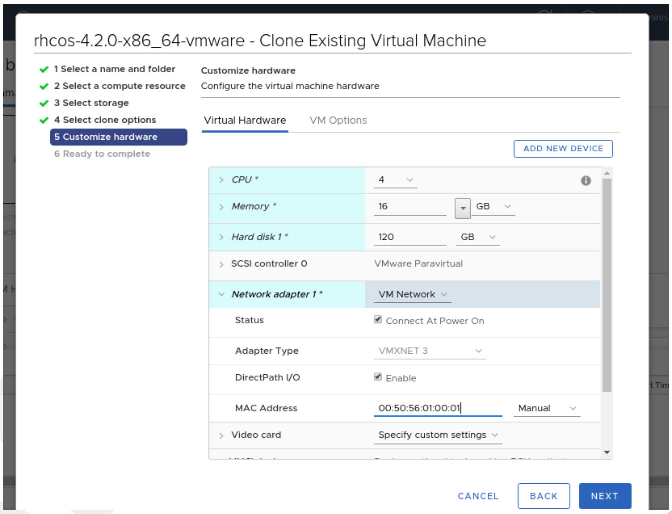

Figure 16. Clone to VM

Click in "**VM Options**" tab and expand "**Advanced**" accordion:

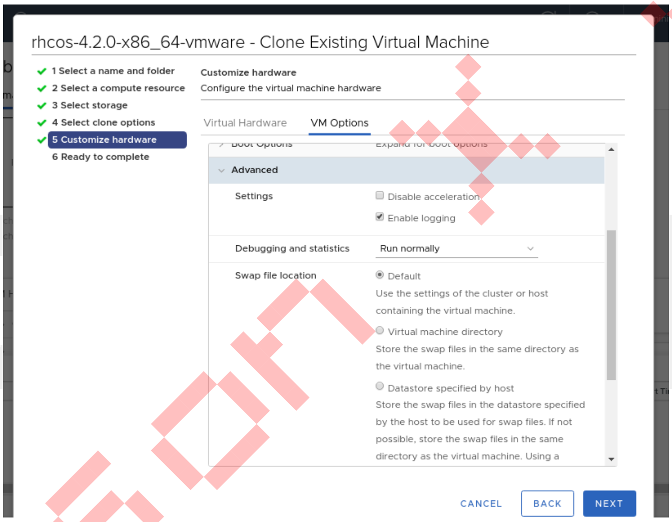

Figure 17. Clone to VM

In **"Latency Sensitivity"** select High and click in "**Edit Configuration...**" button.

Click in the "**ADD CONFIGURATION PARAMS**" button and add the following parameters:

```
guestinfo.ignition.config.data=<content of append_bootstrap.64 file>
guestinfo.ignition.config.data.encoding=base64
disk.EnableUUID=TRUE
```

If using static IP addresses and if 4.14 >= 4.6 set the following parameter before booting the VM:

```
guestinfo.afterburn.initrd.network-kargs=ip=<ipcfg>
```

See: Static IP configuration for vSphere using OVA

> If using static IPs the parameter afterburn.initrd.network-kargs only applies to the first boot.

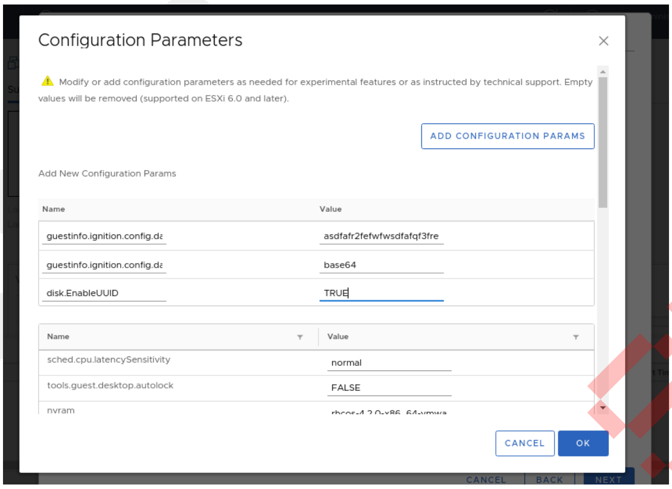

Figure 18. Clone to VM

Click "**NEXT**" then "**FINISH**" to create the bootstrap machine.

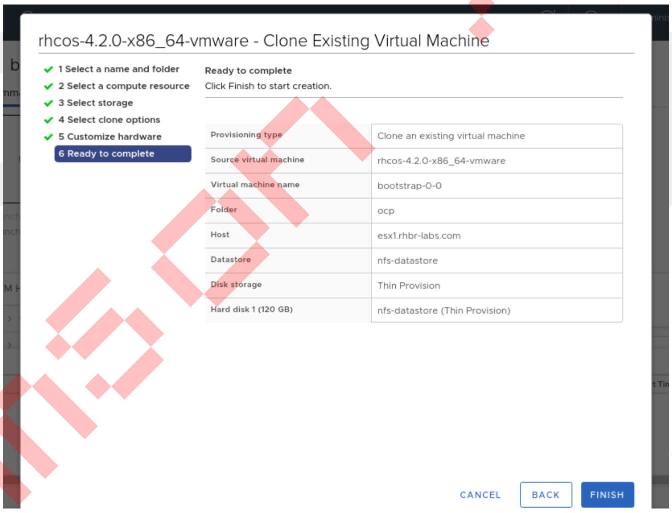

Figure 19. Clone to VM

Repeat the process above to deploy each of the following VMs using the values from this table:

MACHINE	vCPU	RAM	STORAGE	guestinfo.ignition.config.data
master-0	n/a	n/a	n/a	Output of: cat master.64
master-1	n/a	n/a	n/a	Output of: cat master.64
master-2	n/a	n/a	n/a	Output of: cat master.64
worker-0	n/a	n/a	n/a	Output of: cat worker.64
worker-1	n/a	n/a	n/a	Output of: cat worker.64

Once all of the VMs have been deployed, start them up.

C.1.2.8. The Installation Process

See: Waiting for the bootstrap process to complete

The following command was used to install OpenShift with the configuration file created earlier:

```
[user0@infra-services ocp]$ openshift-install wait-for bootstrap-complete --log-level debug
DEBUG OpenShift Installer v4.2.0
DEBUG Built from commit 90ccb37ac1f85ae811c50a29f9bb7e779c5045fb
INFO Waiting up to 30m0s for the Kubernetes API at https://api.ocp.rhbr-labs.com:6443...
INFO API v1.14.6+2e5ed54 up
INFO Waiting up to 30m0s for bootstrapping to complete...
DEBUG Bootstrap status: complete
INFO It is now safe to remove the bootstrap resources
```

After "INFO" message about removing bootstrap resources is displayed, the bootstrap VM and its associated disk can be shut down and removed from vSphere.

The process can take up to 20 minutes. If this message is not displayed within that timeframe, see the troubleshooting tips here: troubleshooting.adoc!

After bootstrap completion, the following command was used to verify the installation:

```
openshift-install wait-for install-complete --log-level debug
```

The output of the command:

```
<Replace this example output with the actual output from the customer environment>

DEBUG OpenShift Installer v4.2.10
DEBUG Built from commit 6ed04f65b0f6a1e11f10afe658465ba8195ac459
INFO Waiting up to 30m0s for the cluster at https://api.ocp.rhbr-labs.com:6443 to initialize...
DEBUG Still waiting for the cluster to initialize: Working towards 4.2.10: 99% complete, waiting on authentication,
console, image-registry
DEBUG Still waiting for the cluster to initialize: Working towards 4.2.10: 99% complete, waiting on authentication,
console, image-registry
DEBUG Still waiting for the cluster to initialize: Working towards 4.2.10: 100% complete
DEBUG Cluster is initialized
INFO Waiting up to 10m0s for the openshift-console route to be created...
DEBUG Route found in openshift-console namespace: console
DEBUG Route found in openshift-console namespace: downloads
DEBUG OpenShift console route is created
INFO Install complete!
INFO To access the cluster as the system:admin user when using 'oc', run 'export
KUBECONFIG=/home/user0/ocp/auth/kubeconfig'
INFO Access the OpenShift web-console here: https://console-openshift-console.apps.ocp.rhbr-labs.com
```

```
INFO Login to the console with user: kubeadmin, password: YsviS-yGfBx-t6FsV-BZ58B
```

C.1.2.9. Running oc Commands

See: Logging in to the cluster

The following command will copy the Kubernetes configuration to the logged-in user's profile and allow the use of the "**oc**" command on the newly deployed cluster:

```
mkdir ~/.kube/
cp auth/kubeconfig ~/.kube/config
```

C.1.2.10. Cluster Operators Deployment

See: Initial Operator configuration

Many operators are deployed as part of the installation process.

The query below shows the operators deployed by the installation process for this engagement:

```
<replace this example with actual output from the customer's environment>

[user0@infra-services ocp]$ watch -n 10 'oc get clusteroperators'
Every 10.0s: oc get clusteroperators
infra-services.rhbr-labs.com: Mon Dec 16 20:43:44 2019

NAME                                       VERSION   AVAILABLE   PROGRESSING   DEGRADED   SINCE
authentication                                       Unknown     Unknown       True   3m1s
cloud-credential                           4.2.10    True        False         False      6m54s
console                                    4.2.10    Unknown     True          False      11s
dns                                        4.2.10    True        False         False      6m20s
image-registry                                       False   False         True   11s
ingress                                    unknown   False   True          False      11s
insights                                   4.2.10    True        False         False      6m53s
kube-apiserver                             4.2.10    True        False         False      4m24s
kube-controller-manager                    4.2.10    True        False         False      4m18s
kube-scheduler                             4.2.10    True        False         False      4m16s
machine-api                                4.2.10    True        False         False      6m56s
machine-config                             4.2.10    True        False         False      6m18s
marketplace                                          False   True          False      12s
monitoring                                           Unknown     True          Unknown    14s
network                                    4.2.10    True        False         False      5m57s
node-tuning                                4.2.10    True        False         False      2m50s
openshift-apiserver                        4.2.10    True        False         False      2m9s
openshift-controller-manager               4.2.10    True        False         False      3m7s
openshift-samples                                    False   False                    9s
operator-lifecycle-manager                 4.2.10    True        False         False      5m52s
operator-lifecycle-manager-catalog         4.2.10    True        False         False      5m52s
operator-lifecycle-manager-packageserver   4.2.10    True        False         False      3m7s
service-ca                                 4.2.10    True        False         False      6m46s
service-catalog-apiserver                  4.2.10    True        False         False      2m57s
```

```
service-catalog-controller-manager          4.2.10    True        False         False      3m
```

C.1.3. Post Deployment Configurations

The point of this section is to configure additional tools that is included with OpenShift. The following actions were performed based on the related documentation:

Section	Configuration	Documentation
Registry Configuration	Set persistent volume and node selector	Configuring the registry for vSphere
Authentication	Set authentication method to HTTPD	Configuring an HTPasswd identity provider
Logging Deployment and Configuration	Deployment and configuration of Logging stack (EFK)	Deploying Cluster Logging Moving the cluster logging resources with node selectors
Monitoring Configuration	Set persistent volume and node selector	Configuring persistent storage Moving monitoring components to different nodes

C.1.3.1. Registry Configuration

Set registry persistent volume:

TODO

C.1.3.2. Authentication

Set authentication with httpd.

```
sudo yum install httpd-tools
htpasswd -c -B -b users.htpasswd admin 'r3dh4t1!'
oc create secret generic htpass-secret --from-file=htpasswd=users.htpasswd -n openshift-config

cat <<EOF > htpasswd-conf.yml
apiVersion: config.openshift.io/v1
kind: OAuth
metadata:
  name: cluster
spec:
  identityProviders:
  - name: my_htpasswd_provider
    mappingMethod: claim
    type: HTPasswd
    htpasswd:
```

```
      fileData:
        name: htpass-secret
EOF

oc apply -f htpasswd-conf.yml
oc adm policy add-cluster-role-to-user cluster-admin admin

oc delete secrets kubeadmin -n kube-system
```

C.1.3.3. Logging Deployment

Deploy: TODO

Set persistent volume: TODO

Set node selector: TODO

C.1.3.4. Monitoring Configuration

Set persistent volume: TODO

Set node selector: TODO

Appendix D: Relevant Links

TODO

Appendix E: Statement of Work

TODO

Main Task

1. Subtask
    a. Detail
    b. Detail

Main Task

1. Subtask
    a. Detail
    b. Detail

Appendix F: Revisions

Table 18. Document Revisions

Version	Date	Changes
0	TODO	• Initial Creation
TODO	TODO	• TODO • TODO

Appendix G: Red Hat Subscriptions Overview

Three factors play an important part in selecting the right RHEL subscription:

- The (Service Level Agreement) SLA
- The number of (physical) Sockets and,
- The number of Virtual Machines.

Red Hat Enterprise Linux can have layered products, like clustering. These require a separate subscription.

G.1. Service Level Agreement

An important part of a subscription is access to Red Hat support. The SLA associated with a subscription plays an important part for the customer here as this defines response times.

	Self-support[1]	Standard	Premium	
Hours of coverage	N/A	Standard business hours	Standard business hours (24x7 for Severity 1 and 2)[2]	
Support channel	None	Web and phone	Web and phone	
Number of cases	N/A	Unlimited	Unlimited	
Response times	**Initial and ongoing response**	**Initial and ongoing response**	**Initial response**	**Ongoing response**
Severity 1	N/A	1 business hour	1 hour	1 hour or as agreed
Severity 2	N/A	4 business hours	2 hours	4 hours or as agreed
Severity 3	N/A	1 business day	4 business hours	8 business hours or as agreed
Severity 4	N/A	2 business day	8 business hours	2 business days or as agreed

Figure 20. Service level agreement response times

G.2. Socket Pairs

A conventional RHEL subscription is based upon socket-pairs. A socket meant here is the *physical* CPU socket.

Any subset of one or two sockets counts for one pair, so a node with 16 cores and two sockets requires one subscription to be compliant. A node with four sockets requires two single socket-pair subscriptions to be compliant.

A node with one socket will then require one subscription to be compliant, and a node with three sockets will require two subscriptions.

Of course, exceptions rule, so a node with four physical sockets of which only two have a seated CPU, will require just one subscription to be compliant.

G.3. Virtualization

A conventional RHEL subscription can alternatively be used to subscribe RHEL virtual machines. The number of Virtual machines included with a subscription is two.

One subscription which would apply to a *physical* dual socket machine can also be used for two Virtual Machines. The number of virtual sockets or vCPUs assigned to the two Virtual Machines being subscribed by a conventional RHEL subscription are not relevant.

G.3.1. Virtual Datacenter Subscriptions

Virtual Datacenter (VDC) Subscriptions differ from conventional RHEL subscriptions in that they subscribe VMs per hypervisor rather than individually.

VDC subscriptions use a parent-child relationship to validate a VM's subscription: the parent (the hypervisor) consumes a subscription while the children (the VMs running on that particular hypervisor) "piggy back" on that subscription. Any supported RHEL VMs running on a hypervisor with a valid VDC subscription have a valid subscription.

Similar to conventional subscriptions, VDC subscriptions are applied to hypervisors per socket-pair. See Socket Pairs.

> A VDC subscription is only valid for the virtual machines running on a hypervisor, not for the hypervisor itself (the hypervisor must have a separate subscription for its own operating system).

G.4. Layered products

Layered products require the same SLA and number of socket-pairs as the RHEL OS these products are deployed upon.

G.5. Red Hat Enterprise Linux life cycle

Red Hat Enterprise Linux Life Cycle

Description	Full Support	Maintenance Support 1 (RHEL 7)[12]	Maintenance Support (RHEL 8 & 9)[13] Maintenance Support 2 (RHEL 7)[12]	Extended Life Phase[7]	Extended Life Cycle Support (ELS) Add-On[8]	Extended Update Support (EUS) Add-On[8] Enahanced Extended Update Support (Enhanced EUS) Add-On[8]
Access to Previously Released Content through the Red Hat Customer Portal	Yes	Yes	Yes	Yes	Yes	Yes
Self-help through the Red Hat Customer Portal	Yes	Yes	Yes	Yes	Yes	Yes
Technical Support[1]	Unlimited	Unlimited	Unlimited	Limited[9]	Unlimited	Unlimited
Asynchronous Security Errata (RHSA)[10 11]	Yes	Yes	Yes	No	Yes[8]	Yes[8]
Asynchronous Bug Fix Errata (RHBA)[2 11]	Yes	Yes	Yes	No	Yes	Yes
Minor Releases	Yes	Yes	Yes (RHEL 7); Not applicable (RHEL 8 & 9)	No	No	No
Refreshed Hardware Enablement[3]	Native	Limited[4] Native	Using Virtualization	Using Virtualization	Using Virtualization	Using Virtualization
Software Enhancements[5]	Yes[6]	No	No	No	No	No
Updated Installation Images	Yes	Yes	Yes[14]	No	No	No

Figure 21. life cycle phases

The Life-cycle has different phases with different types of support. Depending on the applications, the deployment of nodes could be aligned with the different phases; deploy in Full Support, maintain in Maintenance Support 1 and replace in Maintenance Support 2 would be an ideal model.

Life-cycle Dates

All future dates mentioned for "End of Full Support" and "End of Maintenance Support 1" are close approximations, non definitive, and subject to change.

Red Hat Enterprise Linux Life-cycle Dates

Version	General availability	Full support ends	Maintenance Support 1 ends	Maintenance Support or Maintenance Support 2 ends	Extended life cycle support (ELS) add-on ends	Extended life phase ends	Final minor release
Full Support							
9	May 18, 2022	May 31, 2027	Not Applicable	May 31, 2032	May 31, 2035	Ongoing	9.10
8	May 7, 2019	May 31, 2024	Not Applicable	May 31, 2029	May 31, 2032	Ongoing	8.10
Maintenance Support							
7	June 10, 2014	August 6, 2019	August 6, 2020	June 30, 2024	June 30, 2028	Ongoing	7.9
End of maintenance							
7 (System z (Structure A))	April 10, 2018	August 6, 2019	August 6, 2020	May 31, 2021	Not Applicable	Ongoing	7.6
7 (ARM)	November 13, 2017	August 6, 2019	August 6, 2020	November 30, 2020	Not Applicable	Ongoing	7.6
7 (POWER9)	November 13, 2017	August 6, 2019	August 6, 2020	May 31, 2021	Not Applicable	Ongoing	7.6
6	November 10, 2010	May 10, 2016	May 10, 2017	November 30, 2020	June 30, 2024	Ongoing	6.10
5	March 15, 2007	January 8, 2013	January 31, 2014	March 31, 2017	November 30, 2020	Ongoing	5.11

All future dates mentioned are close approximations, non definitive, and subject to change.

Figure 22. Life-cycle dates

The RHEL lifecycle can be found here: https://access.redhat.com/support/policy/updates/errata

Appendix H: Engaging Red Hat Global Support Services

Red Hat® Support helps you get the most from your Red Hat subscriptions. Below, please find information about the Red Hat Customer Portal, Red Hat phone support, your personal points of contact, and how to contact sales and customer service.

We're always available to assist you. If at any time you are unable to reach your personal Red Hat Support contacts, you can find contact information below for the global support leadership team, which is available to assist upon your request.

H.1. Red Hat customer portal: access.redhat.com

The Red Hat Customer Portal is an award-winning digital platform that delivers enterprise product knowledge, subscription resources, and technical expertise that can only come from Red Hat. Use the Customer Portal to access our Knowledgebase of technical expertise, collaborate with peers and Red Hat experts, create, track, and manage your case activity, and much more.

Use of the Customer Portal requires a login/password (linked to your RHN account). If you have any questions about your login and/or password, please contact your <TAM or CSM> if you have assigned.

H.2. Red Hat phone support:

Red Hat technical support engineers are available through phone support to quickly troubleshoot and resolve problems. More information about your regional contact number and business hours can be found at: https://access.redhat.com/support/contact/technicalSupport.

Table 19. RED HAT SUPPORT CONTACTS

TIER	CONTACT NAME & INFORMATION Phone: direct (D) mobile (M)	AVAILABILITY
1	North America Production Support +888-GO-REDHAT (+888-467-3342) https://access.redhat.com/support/[1]	24x7x365 Depending on severity as the first point of contact for Red Hat support.
1	EMEA Production Support +800-GO-REDHAT[1] (+800 4673 3428) https://access.redhat.com/support/[1]	24x7x365 Depending on severity as the first point of contact for Red Hat support.
Additional Global Management Escalation Contacts can be found at: https://access.redhat.com/support/policy/mgt_escalation/		

[1] Use of the Customer Portal requires a Red Hat Network username & password.
[2] Press **3** for "RHEL", then **1** for "Premium Support." Inform the technician that you are with **TODO_customer_portal_account_name**, account number **TODO_customer_portal_account_number**

H.3. Customer service

In addition to Red Hat Support, Red Hat sales and customer service are available to assist you. If you have questions regarding sales, future product needs, or entitlements, please use the contact information provided below.

Table 20. Customer service

CONTACT NAME & INFORMATION	NOTES
Customer Service, North America 888-467-3342	For all questions related to your account, subscriptions, or entitlements, contact customer service, Visit https://access.redhat.com/support/contact/customerService or global contact information. Contact Technical Support for all technical questions or issues related to Red Hat products.
Customer Service, EMEA Germany: 0800 1828065 France (Hors Dom-Tom): 0800907101 France (Dom-Tom): +353 212303445 UK: 0800 032 9515 Italy: 800 979 269 Spain & Portugal: 900 811 831 All other countries: +353 21 2303 445 customerservice-emea@redhat.com	
RED HAT CUSTOMER PORTAL: https://access.redhat.com RED HAT PHONE SUPPORT: 1-888-GO-REDHAT (1-888-467-3342)	RECOMMENDATIONS FOR OPTIMAL SUPPORT EXPERIENCE: https://access.redhat.com/articles/280093 TO ESCALATE A SUPPORT CASE: https://access.redhat.com/support/escalation

H.4. Production support terms of service

https://access.redhat.com/support/offerings/production/sla

	Self-support	Standard	Premium
Hours of coverage	N/A	Standard business hours [2]	Standard business hours (24x7 for Severity 1 and 2) [3]
Support channel	None	Web and phone	Web and phone

	Self-support	Standard	Premium	
Number of cases	N/A	Unlimited	Unlimited	
Response times	Initial and ongoing response	Initial and ongoing response	Initial response	Ongoing response
Severity 1	N/A	1 business hour	1 hour	1 hour or as agreed
Severity 2	N/A	4 business hours	2 hours	4 hours or as agreed
Severity 3	N/A	1 business day	4 business hours	8 business hours or as agreed
Severity 4	N/A	2 business day	8 business hours	2 business days or as agreed

H.5. Red Hat Support severity level definitions

https://access.redhat.com/support/policy/severity/

Opening a support case online can make it easier to share technical data, error messages, and system information with your Red Hat Support representative. Following the online submission with a phone call can reduce response time as well as potential errors in the capture of information.

Red Hat recommends that you follow any Severity 1 and 2 online support case submissions with a phone call to your local support center.

Red Hat Global Support Services uses the following definitions to classify issues:

H.5.1. Severity 1 (urgent)

A problem that severely impacts your use of the software in a production environment (such as loss of production data or in which your production systems are not functioning). The situation halts your business operations and no procedural workaround exists.

H.5.2. Severity 2 (high)

A problem where the software is functioning but your use in a production environment is severely reduced. The situation is causing a high impact to portions of your business operations and no procedural workaround exists.

H.5.3. Severity 3 (medium)

A problem that involves partial, non-critical loss of use of the software in a production environment or development environment. For production environments, there is a medium-to-low impact on your business, but your business continues to function, including by using a procedural workaround. For development environments, where the situation is causing your project to no longer continue or migrate into production.

H.5.4. Severity 4 (low)

A general usage question, reporting of a documentation error, or recommendation for a future product enhancement or modification. For production environments, there is low-to-no impact on your business or the performance or functionality of your system. For development environments, there is a medium-to-low impact on your business, but your business continues to function, including by using a procedural workaround.

H.6. Tips to create a good support case

1. Fill out the account number, your name, it is recommended to be individual instead global account.
2. Select the product that it is affected.
3. Version of the product.
4. Problem statement should be precise to see if they are any KCS that help you on the resolution. Please read the KCS suggested before open a case. Remember use the name of the component that you are having issue - Ceph .... , Nova .... Keywords are very important.
5. Explain your issue with the maximum details and add the commands that you are using with outputs.
6. Problem statement should be descriptive like: [Component][Region] Issue description.
   Examples would be:
   [Nova][NL Region] Nova fails on one compute
   [Ceph][Singapore Region] Ceph without OSD
7. Explain your issue with the maximum details and add the commands that you are using with outputs.

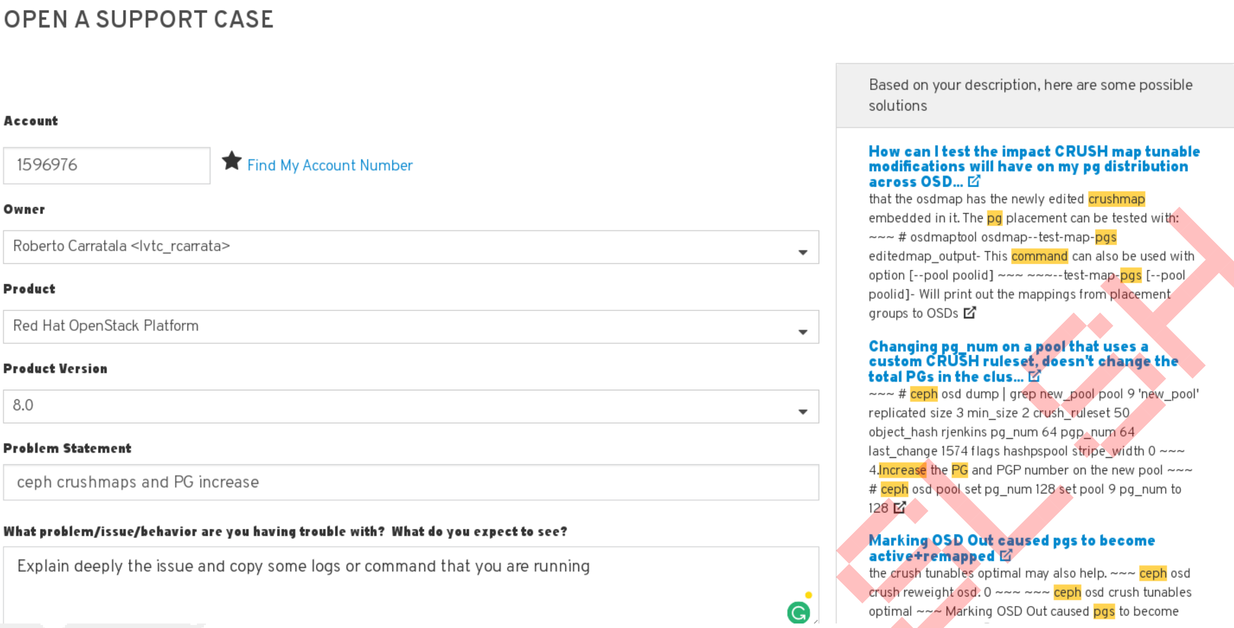

Figure 23. Portal: case

1. It is important to put the environment that it is affected, pre production, production and what region is affecting, London, New York
   Remember the severity levels and assign it properly.
2. Do attach sosreports of the servers/components affect for speedup the case: https://access.redhat.com/solutions/3592
3. Select server of support, add premium any time that you have the option.

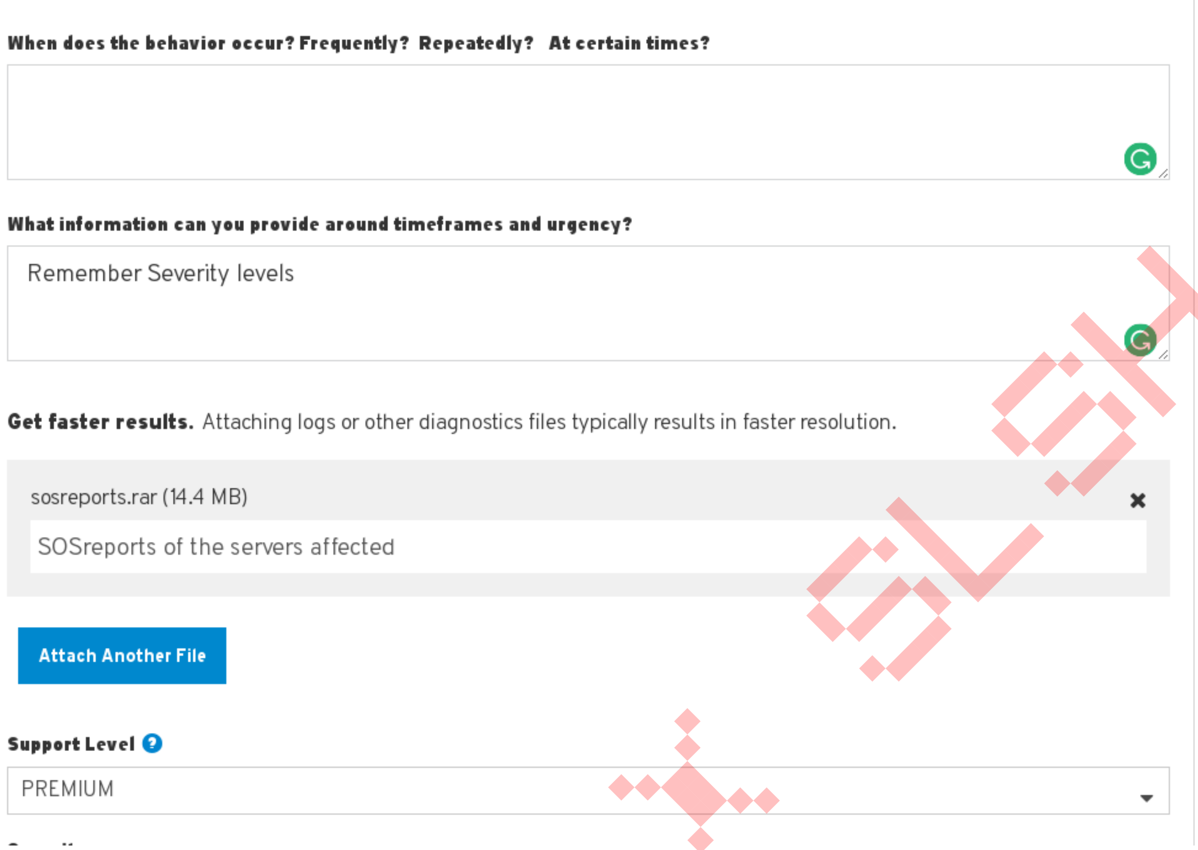

Figure 24. Portal: case details

1. Select severity level.
2. Add into notification to someone or distribution list.
3. If available, select a case group to notify other member, per example Cloud, alert all member of Cloud from your team and they could work on the case too.

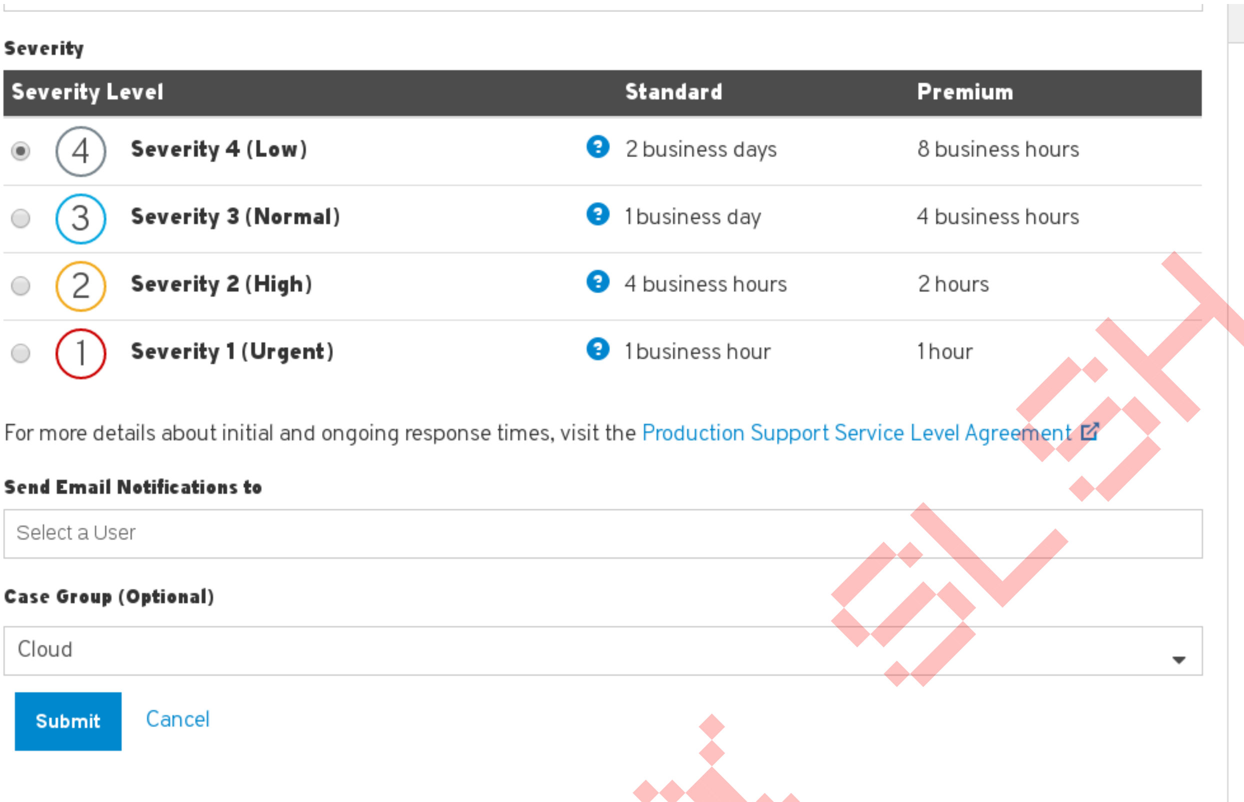

Figure 25. Portal: set severity and add notification